Force Control and Human Interaction with the FR3 manipulator

Danai Chanlaridou

University of Ioannina

Department of Computer Science and Engineering

Supervisor: Konstantinos Vlachos



Presentation roadmap

1. Motivation, Goals, Problem Statement
2. Franka Research 3 Overview
3. Mathematical model
4. Force Control Algorithms
5. Experiments & results
6. Conclusions

Motivation, Goals, Problem Statement

Why force-aware interaction matters



Motivation, Goals, Problem Statement

From rigid automation to safe physical interaction



Motivation

To enable the FR3 manipulator to interact safely and compliantly with humans and the environment during contact-rich tasks, including physiotherapy rehabilitation applications.



Goal

To evaluate how the FR3 behaves as a force-aware collaborative robot during contact and human interaction.



Problem Statement

Traditional position control is not sufficient for physical interaction, because small position errors during contact can generate large and unsafe forces.



Thesis focus

Study, implement, and evaluate the force-control methods on the Franka Research 3 on real hardware.

Motion + Force + Safety

Collaborative Manipulation

Franka Research 3 Overview

Platform overview and key capabilities



Franka Research 3 Overview

Franka Research 3 (FR3) is a 7-DOF collaborative robotic manipulator designed for research, education, and human-robot interaction.

Key characteristics

- 7 revolute joints
- Redundant robotic structure
- 3 kg payload
- 855 mm maximum reach
- Integrated joint torque sensors in all 7 axes
- Torque-level control through Franka Control Interface
- Suitable for force control and physical human–robot interaction



Mathematical Model

$$\dot{x} = f(q)$$
$$\dot{x} = J(q)\dot{q}$$
$$\tau = J^T(q)F$$



Forward / inverse kinematics

Mapping between joint space (q) and Cartesian space (x).



Jacobian matrix

Differential relationship between joint velocities and end-effector linear/angular velocities.



$$M(q)\ddot{q} + C(q,\dot{q}) + g(q) + F_{ext} + F_{fric} = \tau$$

Dynamic model

Relationship between joint torques, inertia, Coriolis/centrifugal effects, gravity, external forces, and friction in robot motion.



Forward Kinematics

Compute the end-effector pose from the seven joint angles

1) Joint configuration

The configuration of the 7-DOF manipulator is represented as a 7-dimensional vector.

$$q = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 & q_7 \end{bmatrix}^T \in \mathbb{R}^7$$

2) End-effector pose

The end-effector pose is represented by a homogeneous transformation matrix containing both position and orientation.

$${}^0T_e(q) = \begin{bmatrix} {}^0R_e(q) & {}^0p_e(q) \\ 0 & 1 \end{bmatrix} \in SE(3)$$

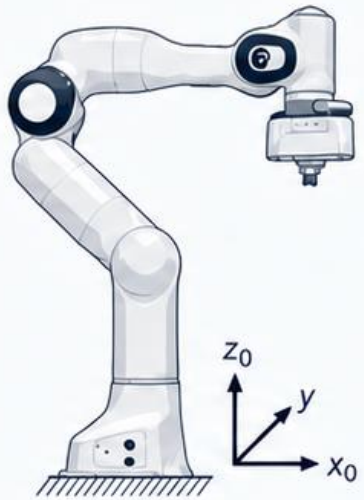
3) Serial chain composition

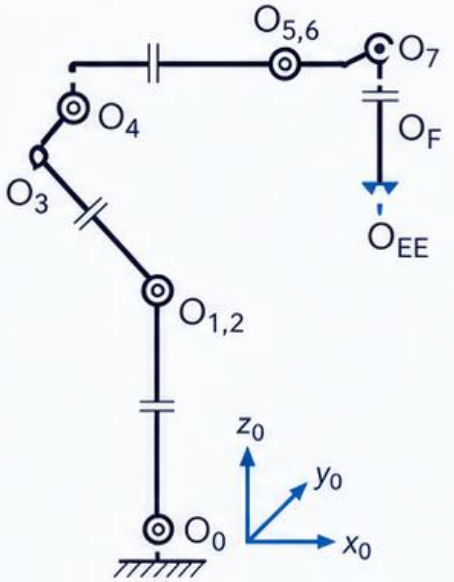
The final pose is obtained by multiplying the transformations from the base frame to the end-effector frame.

$${}^0T_e(q) = T_1(q_1)\, T_2(q_2) \cdots T_7(q_7)\, T_e$$

Key idea: forward kinematics maps joint space to Cartesian pose.

FR3
7-DOF Manipulator





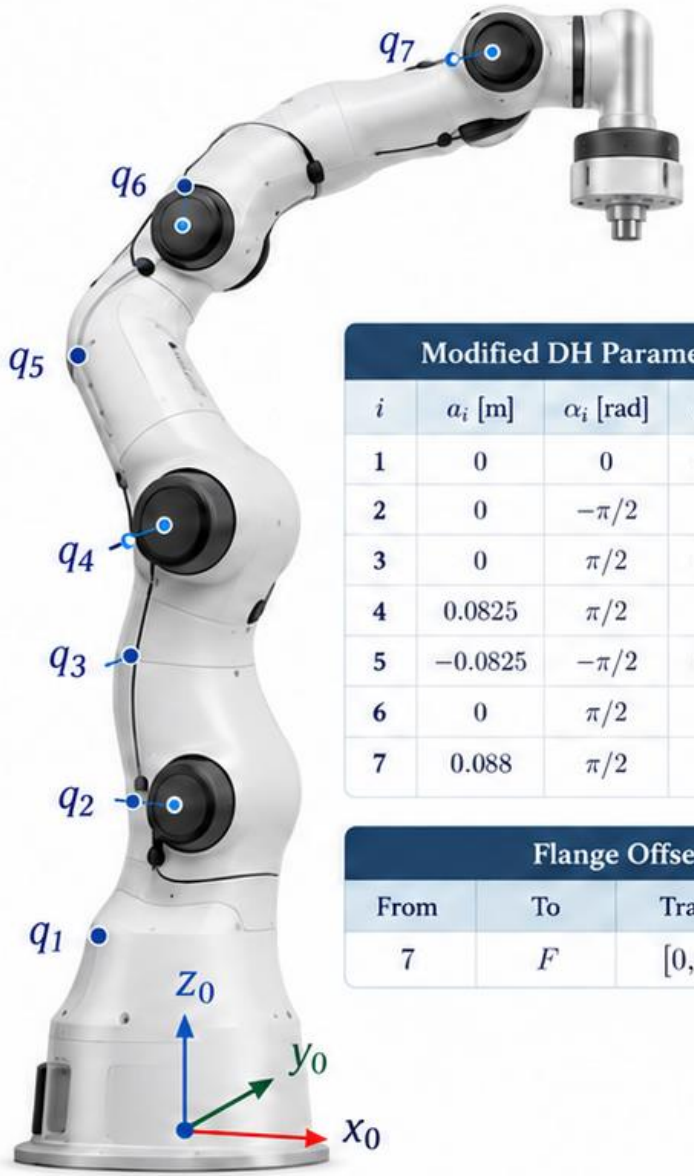
Notation and symbols

q	:	joint configuration vector (7×1)
q_i	:	i-th joint angle $(i = 1, \ldots, 7)$
${}^0T_e(q)$	:	end-effector homogeneous transform relative to base frame
${}^0R_e(q)$	:	end-effector rotation matrix (3×3)
${}^0p_e(q)$	:	end-effector position vector (3×1)
$SE(3)$	:	space of rigid-body transforms
$T_i(q_i)$	:	transform of link/joint i
T_e	:	tool/end-effector transform (from flange to end-effector)

O_i	:	origin of frame i (joint i)
O_0	:	base frame origin
$O_{1,2}$	:	origin at joints 1 and 2 (coincident in diagram)
O_3	:	origin of joint 3
O_4	:	origin of joint 4
$O_{5,6}$	:	origin at joints 5 and 6 (coincident in diagram)
O_7	:	origin of joint 7
O_F	:	flange (tool) frame origin
O_{EE}	:	end-effector frame origin
x_0, y_0, z_0	:	base-frame axes

Homogeneous Transformations for the FR3 Manipulator

Modified Denavit–Hartenberg (Craig convention)



Modified DH Parameters (FR3)				
i	a_i [m]	α_i [rad]	d_i [m]	θ_i
1	0	0	0.333	q_1
2	0	$-\pi/2$	0	q_2
3	0	$\pi/2$	0.316	q_3
4	0.0825	$\pi/2$	0	q_4
5	-0.0825	$-\pi/2$	0.384	q_5
6	0	$\pi/2$	0	q_6
7	0.088	$\pi/2$	0	q_7

Flange Offset		
From	To	Translation [m]
7	F	$[0, 0, 0.107]^T$

1. Joint 1: ${}^0T_1$

$$ {}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0.333 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

2. Joint 2: ${}^1T_2$

$$ {}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

3. Joint 3: ${}^2T_3$

$$ {}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ 0 & 0 & -1 & -0.316 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

4. Joint 4: ${}^3T_4$

$$ {}^3T_4 = \begin{bmatrix} c_4 & -s_4 & 0 & 0.0825 \\ 0 & 0 & -1 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

5. Joint 5: ${}^4T_5$

$$ {}^4T_5 = \begin{bmatrix} c_5 & -s_5 & 0 & -0.0825 \\ 0 & 0 & 1 & 0.384 \\ -s_5 & -c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

6. Joint 6: ${}^5T_6$

$$ {}^5T_6 = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

7. Joint 7: ${}^6T_7$

$$ {}^6T_F = \begin{bmatrix} c_7 & -s_7 & 0 & 0.088 \\ 0 & 0 & -1 & 0 \\ s_7 & c_7 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

8. Flange / End-effector: ${}^7T_F$

$$ {}^7T_F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.107 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Overall Homogeneous Transformation (Base to End-Effector)

$$ {}^0T_F = {}^0T_1 \; {}^1T_2 \; {}^2T_3 \; {}^3T_4 \; {}^4T_5 \; {}^5T_6 \; {}^6T_7 \; {}^7T_F \longrightarrow {}^0T_F = \begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Inverse Kinematics

Compute joint configurations for a desired Cartesian pose

Goal (desired end-effector pose)

$$^{0}T_e(q) = T_d$$



1 Local linearization

$$\Delta x \approx J(q)\Delta q$$

Relate a small change in Cartesian pose Δx to a small change in joint variables Δq.

2 Damped least-squares solution

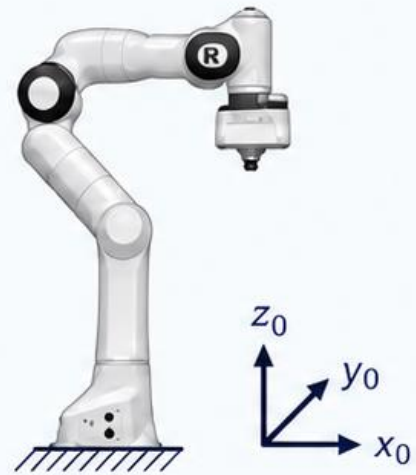
$$\Delta q = J^{\dagger}(q)\Delta x$$

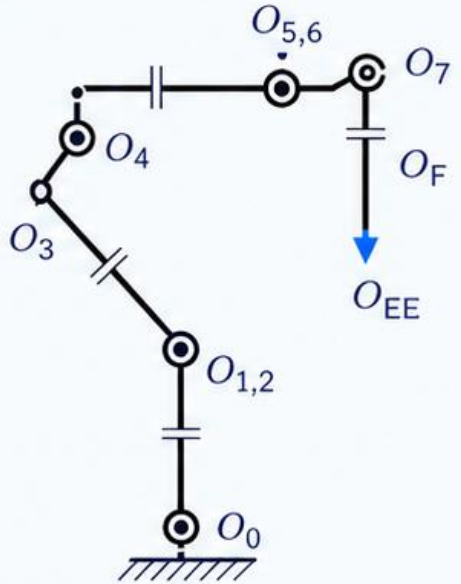
$$\dot{q} = J^{T}(JJ^{T} + \lambda^{2} I)^{-1} v$$

Compute a stable joint increment using the damped least-squares (DLS) inverse of the Jacobian.

Key idea: inverse kinematics maps a desired Cartesian pose back to joint space.

FR3 7-DOF Manipulator





Notation

$^{0}T_e(q)$	:	End-effector pose w.r.t. base frame
T_d	:	Desired end-effector pose
Δx	:	Small change in Cartesian pose
$J(q)$	:	Geometric Jacobian at q
Δq	:	Small change in joint variables
q	:	Joint configuration (7×1)
J^T	:	Jacobian transpose
λ	:	Damping factor (scalar)
I	:	Identity matrix (6×6)
v	:	Desired Cartesian velocity (6×1)

O_0	Base frame (joint 0)
$O_{1,2}$	Frame at joints 1–2
O_3	Frame at joint 3
O_4	Frame at joint 4
$O_{5,6}$	Frame at joints 5–6
O_7	Frame at joint 7
O_F	Flange frame
O_{EE}	End-effector frame
x_0	Base frame x-axis
y_0	Base frame y-axis
z_0	Base frame z-axis

Jacobian Matrix

1 Motion mapping: joint velocities → Cartesian twist

$$v = J(q)\dot{q}$$

v : Cartesian twist (6×1)
$\dot{q}$: joint velocities (7×1)
$J(q)$: geometric Jacobian (6×7)

Joint velocity vector

$$\dot{q} = \begin{bmatrix} \dot{q}_1 & \dot{q}_2 & \dot{q}_3 & \dot{q}_4 & \dot{q}_5 & \dot{q}_6 & \dot{q}_7 \end{bmatrix}^T \in \mathbb{R}^7$$

Cartesian twist

$$v = \begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \begin{matrix} \} \\ \} \end{matrix} \begin{matrix} \text{Linear velocity } [m/s] \\ \text{Angular velocity } [rad/s] \end{matrix}$$

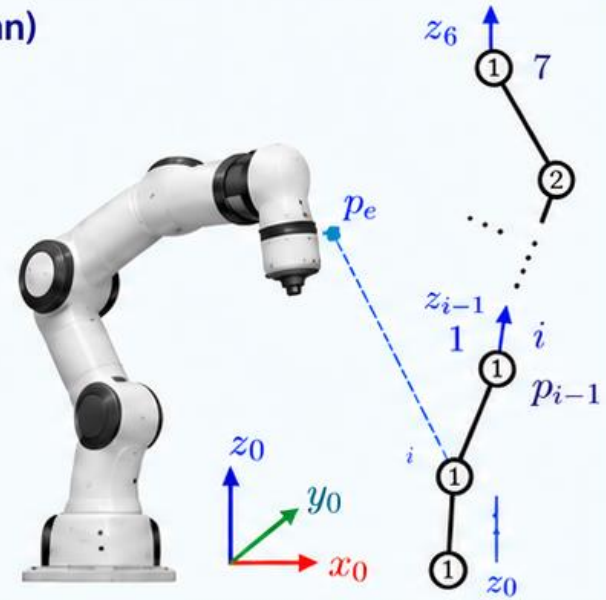
2 Jacobian structure (i-th column)

$$J(q) = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix}$$

The i-th column of $J(q)$ is given by

Revolute joint

$$J_{vi} = z_{i-1} \times (p_e - p_{i-1})$$
$$J_{\omega i} = z_{i-1}$$



3 Force mapping: Cartesian wrench → joint torques

$$\tau = J^T(q)F$$

F : Cartesian wrench (6×1)
τ : joint torques (7×1)
$J(q)$: geometric Jacobian (6×7)

Cartesian wrench $F = \begin{bmatrix} F_x & F_y & F_z & M_x & M_y & M_z \end{bmatrix}^T \in \mathbb{R}^6$

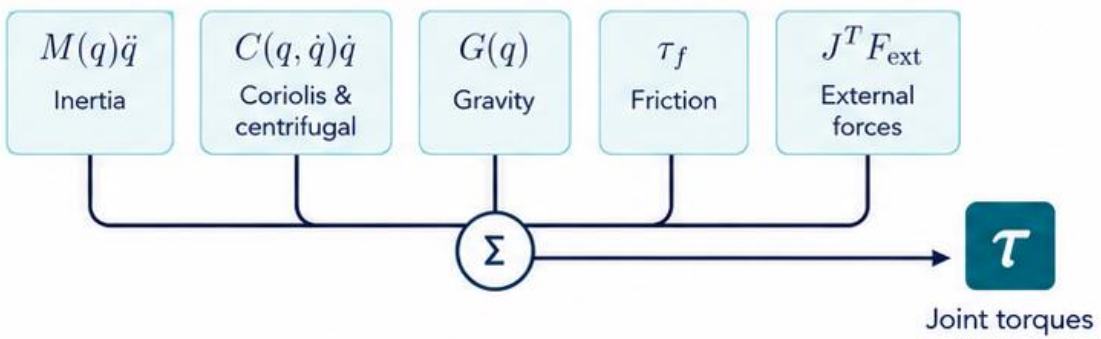
Joint torques $\tau = \begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 & \tau_5 & \tau_6 & \tau_7 \end{bmatrix}^T \in \mathbb{R}^7$

Dynamic Model for Force Control

Model terms and inverse dynamics for safe torque control

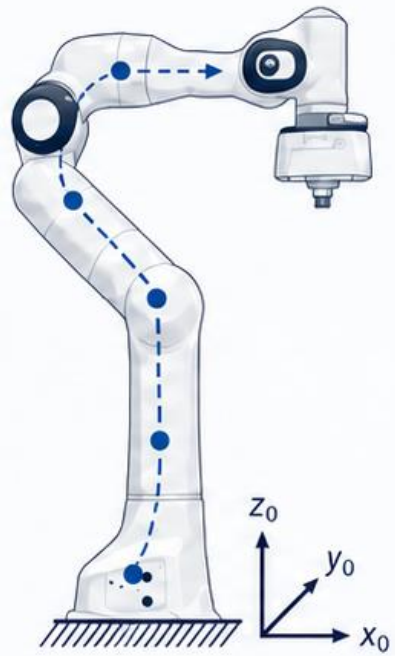
$$\tau = \underbrace{M(q)\ddot{q}}_{\text{Inertia}} + \underbrace{C(q,\dot{q})\dot{q}}_{\text{Coriolis \& centrifugal}} + \underbrace{G(q)}_{\text{Gravity}} + \underbrace{\tau_f}_{\text{Friction}} + \underbrace{J^T F_{\text{ext}}}_{\text{External forces}}$$

Model terms contribute to joint torques



RNEA: Recursive Newton-Euler Algorithm

Efficient inverse dynamics via two recursive passes



1 Forward pass

Compute link velocities and accelerations from base to end-effector.

2 Backward pass

Propagate forces and moments toward the base from end-effector to base.

Output: joint torques τ required to realize desired motion and external forces.

Force control algorithms

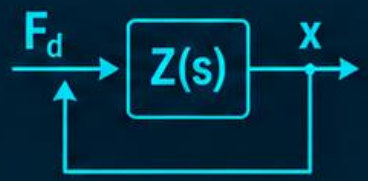
01



Joint Position Baseline

Tracks a desired joint position trajectory as a stable baseline for force control.

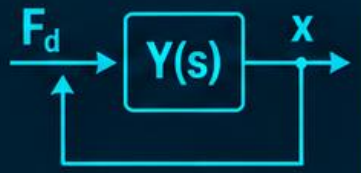
02



Cartesian Impedance Control

Shapes dynamic interaction in task space using impedance models for accurate force regulation.

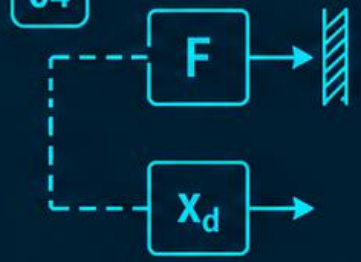
03



Cartesian Admittance Control

Converts measured forces into motion commands using admittance models for compliant behavior.

04



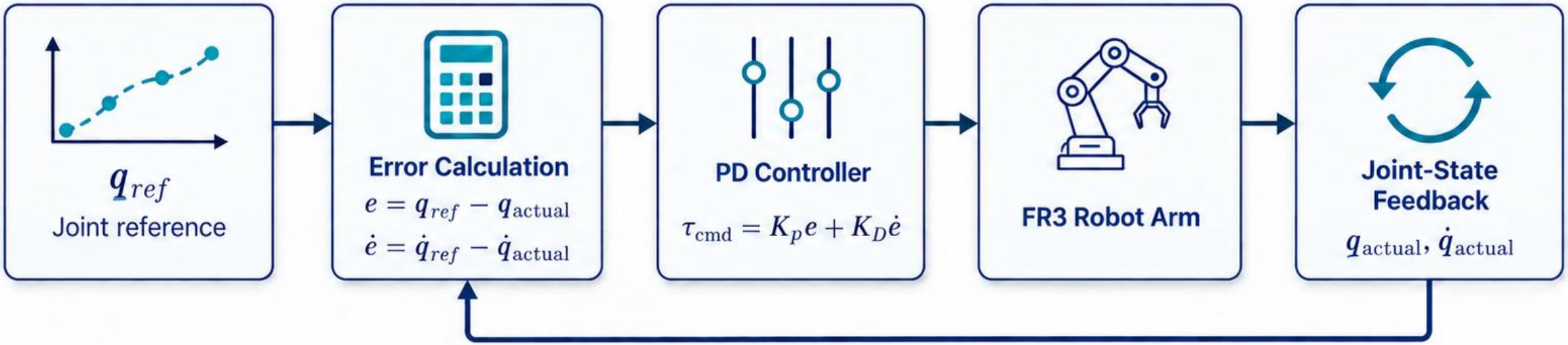
Hybrid Force/Position Control

Combines force control in selected directions with position control in the remaining directions.



Joint Position Baseline Control

Reference joint trajectory tracking in free space



Notation

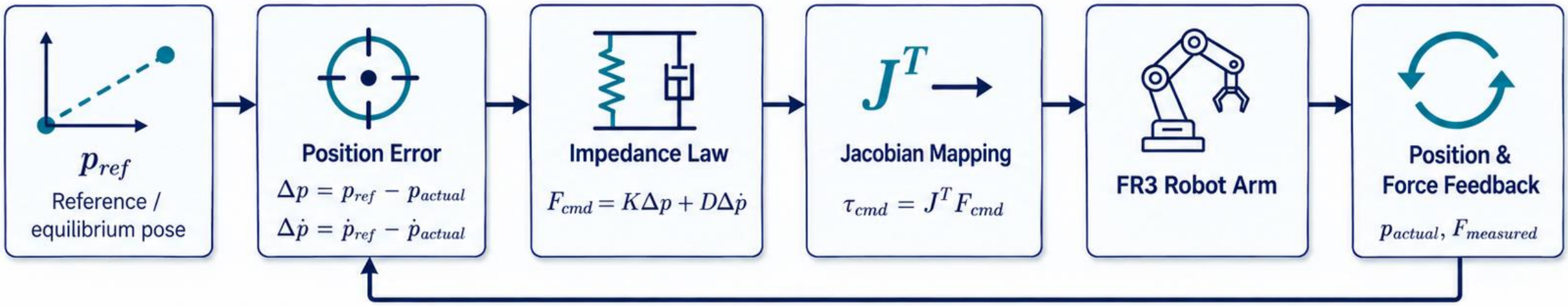
q_{ref}	Reference joint position (desired trajectory)
q_{actual}	Actual joint position (measured)
$\dot{q}_{ref}$	Reference joint velocity
$\dot{q}_{\text{actual}}$	Actual joint velocity (measured)

e	Position tracking error ($e = q_{ref} - q_{\text{actual}}$)
$\dot{e}$	Velocity tracking error ($\dot{e} = \dot{q}_{ref} - \dot{q}_{\text{actual}}$)
K_p	Proportional gain
K_D	Derivative gain
τ_{cmd}	Commanded joint torque

τ	Applied joint torque ($\tau = \tau_{\text{cmd}}$)
q	Joint position vector ($q = [q_1 \quad q_2 \quad \cdots \quad q_7]^T$)
$\dot{q}$	Joint velocity vector ($\dot{q} = [\dot{q}_1 \quad \dot{q}_2 \quad \cdots \quad \dot{q}_7]^T$)

Cartesian Impedance Control

Motion error produces a force response



$$F_{cmd} = K\left(p_{ref} - p_{actual}\right) + D\left(\dot{p}_{ref} - \dot{p}_{actual}\right)$$

Notation

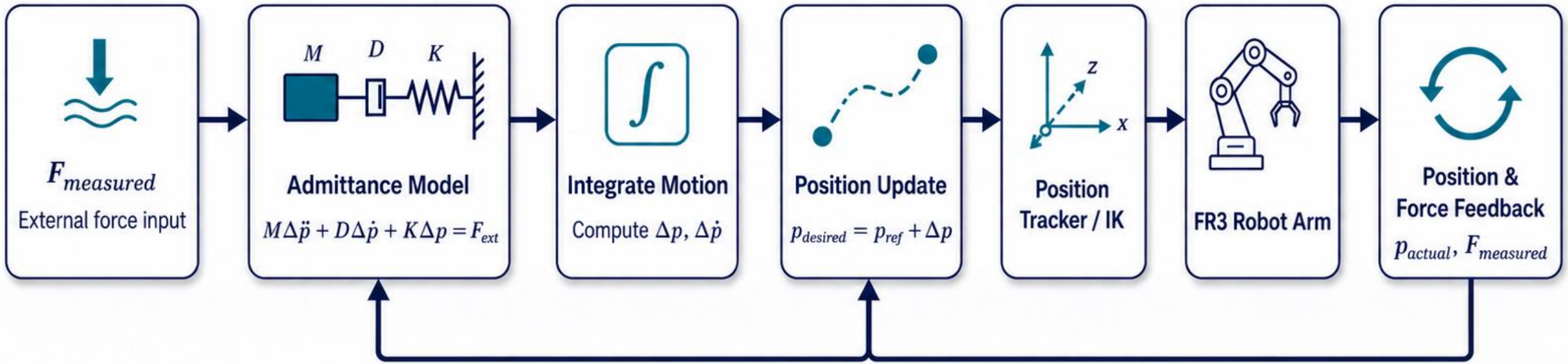
p_{ref}	Reference / desired Cartesian position (equilibrium pose)
p_{actual}	Actual Cartesian position (measured)
Δp	Position error $(\Delta p = p_{ref} - p_{actual})$
$\dot{p}_{ref}$	Reference Cartesian velocity (desired, usually zero)
$\dot{p}_{actual}$	Actual Cartesian velocity (measured)
$\Delta \dot{p}$	Velocity error $(\Delta \dot{p} = \dot{p}_{ref} - \dot{p}_{actual})$

F_{cmd}	Commanded Cartesian force (from impedance law)
K	Cartesian stiffness matrix (typically 6×6)
D	Cartesian damping matrix (typically 6×6)
τ_{cmd}	Commanded joint torques $(\tau_{cmd} = J^T F_{cmd})$
J	Jacobian matrix (maps joint velocities to Cartesian velocities)
J^T	Jacobian transpose (maps Cartesian forces to joint torques)

τ	Joint torque vector $(\tau = [\tau_1 \ \tau_2 \ \dots \ \tau_7]^T)$
p_{actual}	Actual Cartesian position (measured)
$F_{measured}$	Measured Cartesian force (e.g., from force/torque sensor)
x	Cartesian position $(x = [x \ y \ z]^T)$
$\dot{x}$	Cartesian velocity $(\dot{x} = [\dot{x} \ \dot{y} \ \dot{z}]^T)$
F	Cartesian force vector $(F = [F_x \ F_y \ F_z]^T)$

Cartesian Admittance Control

External force produces compliant motion



$$M\Delta\ddot{p} + D\Delta\dot{p} + K\Delta p = F_{ext}$$

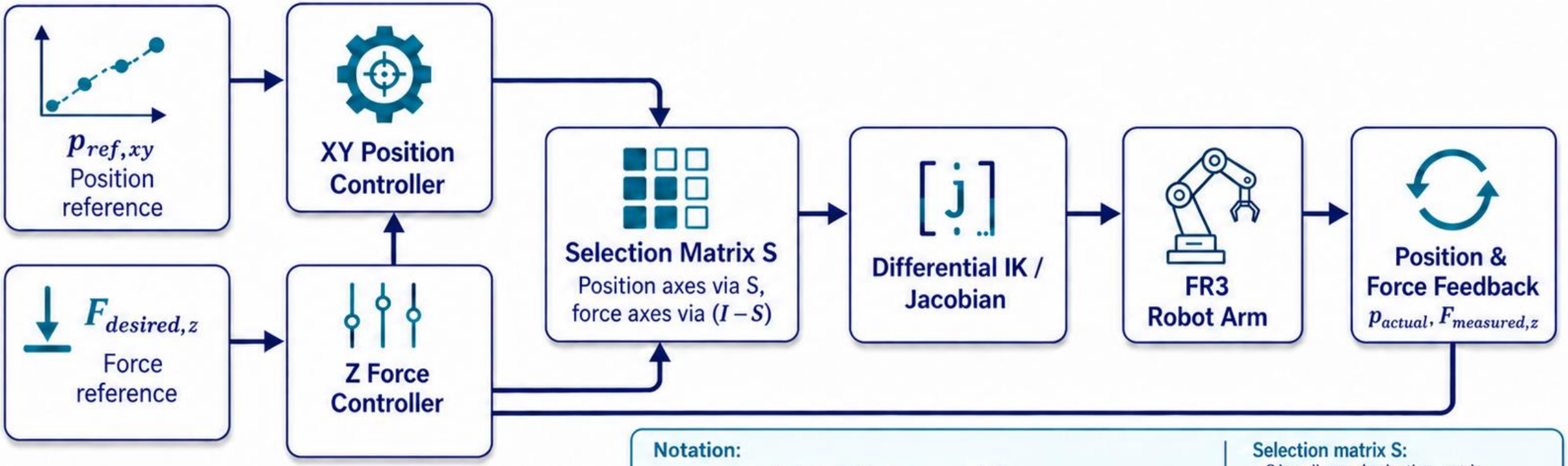
Notation

$F_{measured}$	:	measured external Cartesian force
F_{ext}	:	external force input to the admittance model
M	:	virtual inertia matrix
D	:	virtual damping matrix
K	:	virtual stiffness matrix
Δp	:	Cartesian position displacement
$\Delta\dot{p}$	:	Cartesian velocity displacement
$\Delta\ddot{p}$	:	Cartesian acceleration displacement
p_{ref}	:	reference / equilibrium Cartesian position
$p_{desired}$	:	desired Cartesian position after admittance update
p_{actual}	:	measured Cartesian position

Position Tracker / IK : maps desired Cartesian motion to robot commands (inverse kinematics)

Hybrid Force / Position Control

Position in selected axes and force in selected axes



Notation:

$p_{ref,xy}$	:	desired position in x, y axes [m]
$F_{desired,z}$	:	desired force in z axis [N]
p_{actual}	:	measured end-effector position [m]
$F_{measured,z}$	:	measured force in z axis [N]
J	:	Jacobian matrix (used for differential inverse kinematics)
Differential IK	:	computes joint velocities from task-space velocity, $\dot{q} = J^{-1}v$

Selection matrix S:

S is a diagonal selection matrix that chooses position-controlled axes and force-controlled axes.

$S_{ii} = 1$ → axis i is position-controlled

$S_{ii} = 0$ → axis i is force-controlled

$(I-S)$ selects the complementary force-controlled axes.

Example:

$S = \mathrm{diag}(1, 1, 0)$

controls position in x, y and force in z.

Experiments

1 

1. Pick-and-place demo

2 

2. Human push / disturbance rejection

3 

3. Hand-guiding / lead-through

4 

4. Hybrid Position/Force Interaction

5 

5. Physiotherapy interaction shoulder

6 

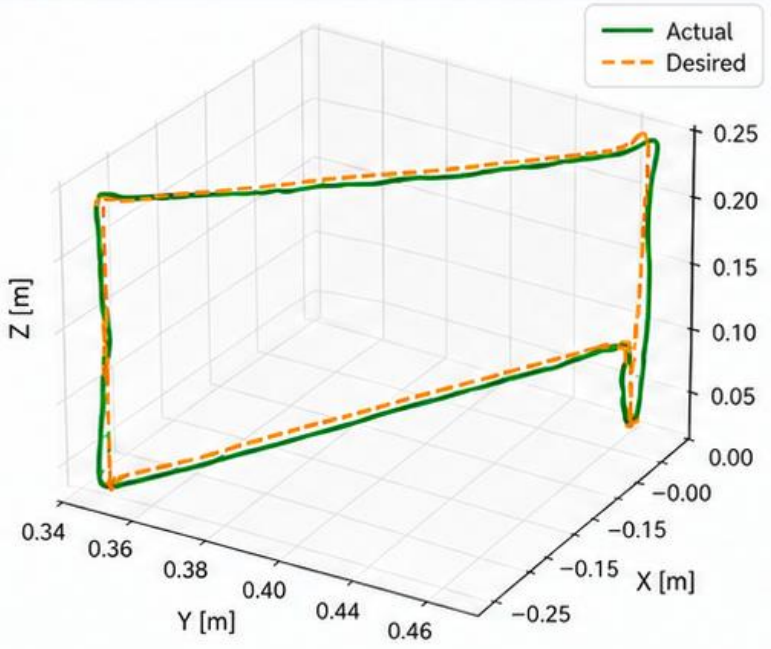
6. Physiotherapy interaction elbow



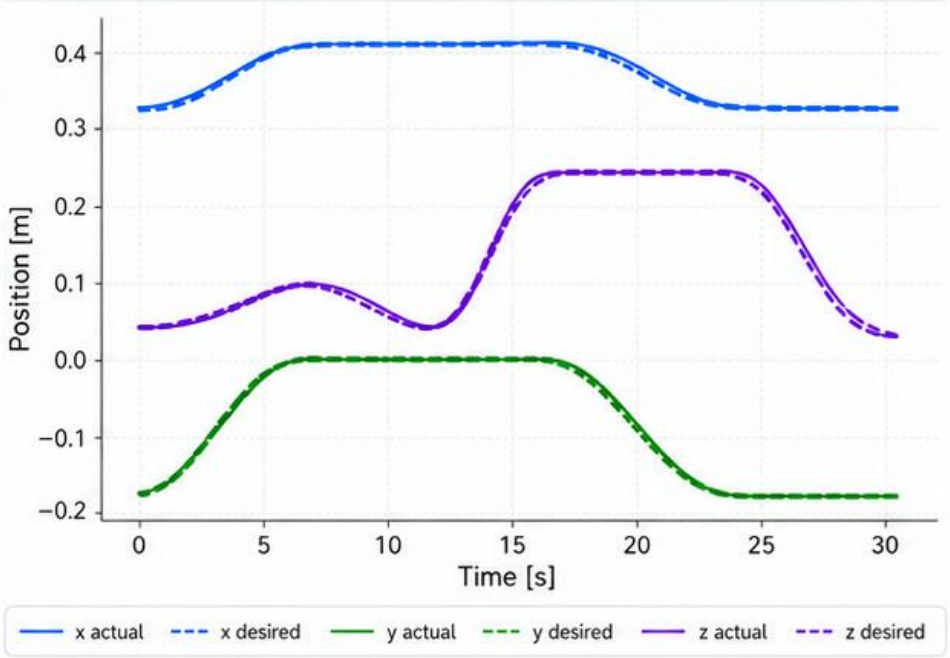
1. Pick & Place Demo

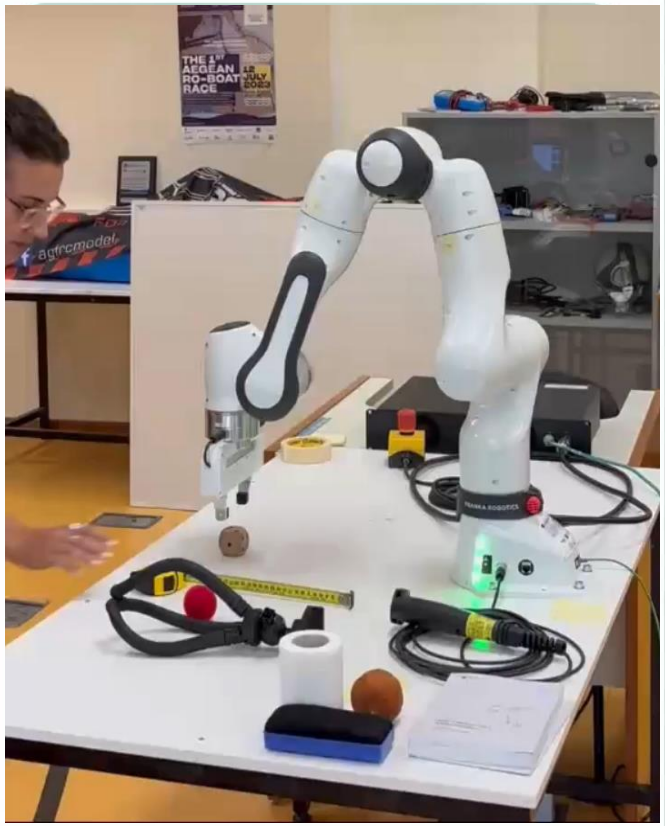
Demonstrating accurate path tracking and smooth motion during object transfer.

3D End-Effector Path



Cartesian Position vs Time

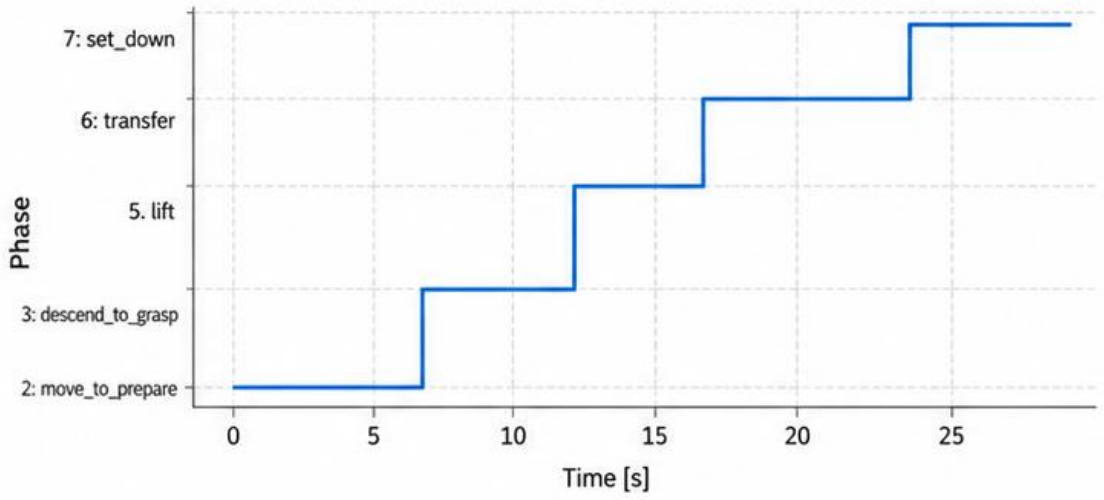




Pick & Place Demo

1 Pick-and-Place Phase Timeline

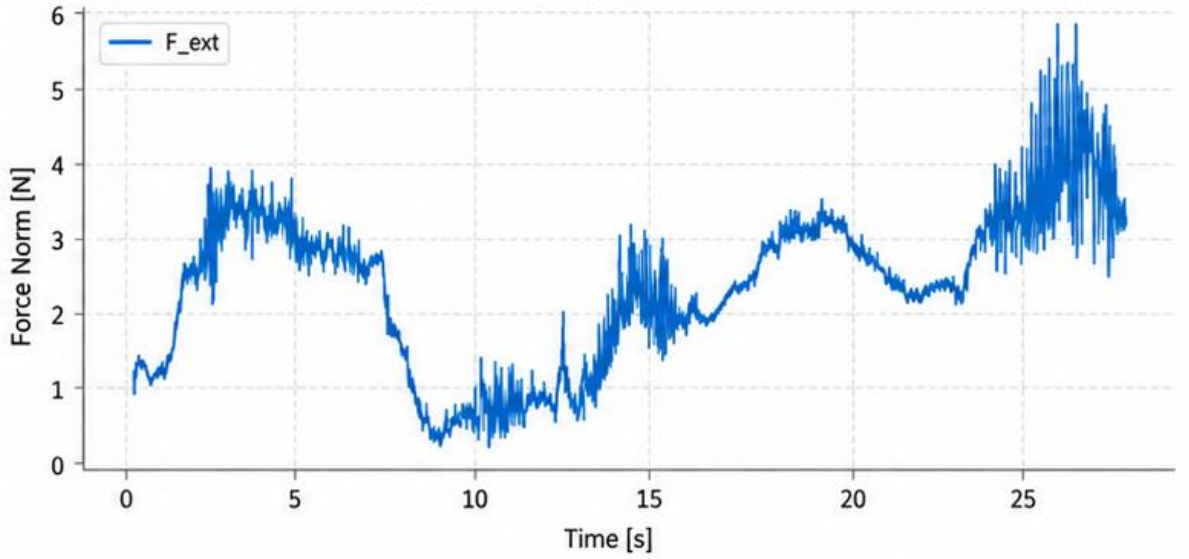
Sequence of task phases over time.



2	move_to_prepare	Robot moves to the pre-grasp pose.	6	transfer	Robot moves the object to the target.
3	descend_to_grasp	Robot descends to grasp the object.	7	set_down	Robot places the object down.
5	lift	Robot lifts the object.			

2 External Force Norm

Measured interaction force magnitude during the task.



Peak Force	Mean Force	Steady Control	Safe Interaction
5.8 N	2.4 N	Low Force Oscillations	Forces within safe limits

Human Push and Disturbance Rejection

Comparison of external joint torque and end-effector displacement under soft and stiff control.

SOFT CONTROLLER

Compliant behavior





The soft controller allows compliant interaction, resulting in larger displacements under human pushes and effective disturbance rejection.

STIFF CONTROLLER

Stiff behavior





The stiff controller maintains small displacements under human pushes, resulting in stronger disturbance rejection and higher interaction forces.

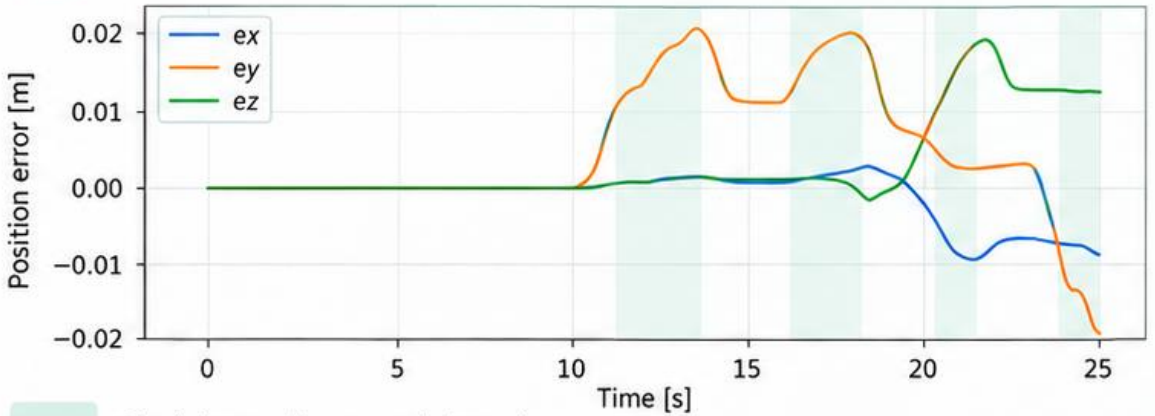
Human Push and Disturbance Rejection

Soft version

Stiff version

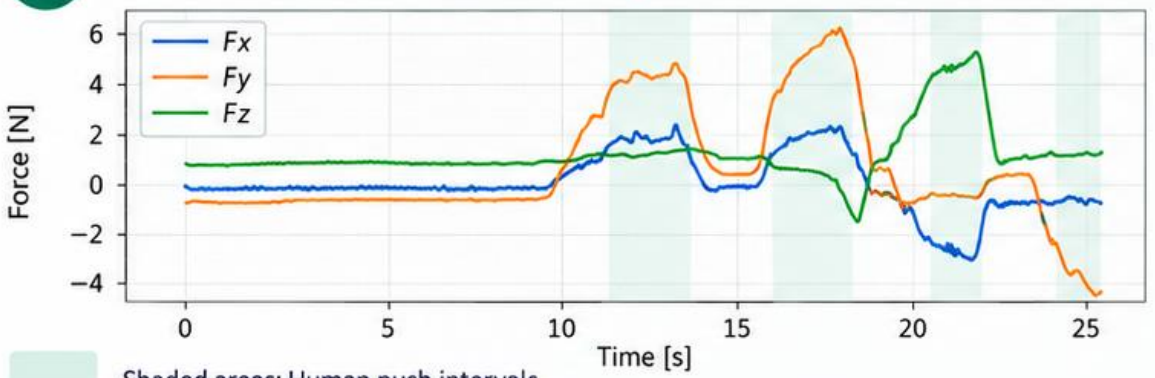
Comparison of the robot's response to human pushes using soft and stiff controllers.
The soft controller results in larger displacement/position errors under smaller forces,
while the stiff controller maintains smaller errors under larger forces.

Cartesian Position Error Components



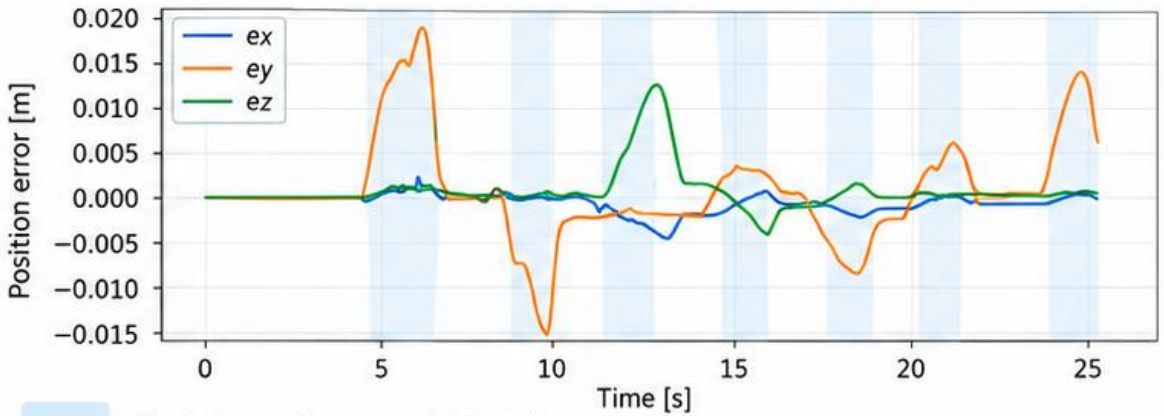
Shaded areas: Human push intervals

Estimated External Force Components



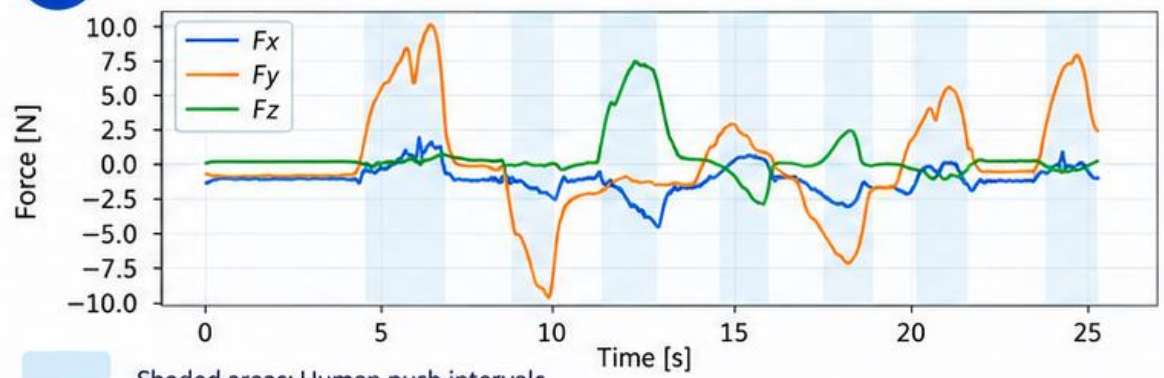
Shaded areas: Human push intervals

Cartesian Position Error Components



Shaded areas: Human push intervals

Estimated External Force Components



Shaded areas: Human push intervals

Human Push and Disturbance Rejection

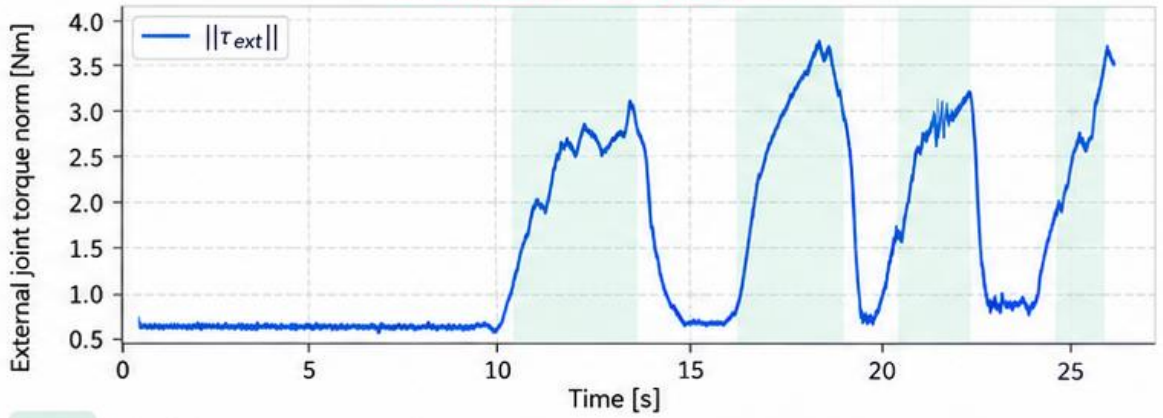
Soft version

Stiff version

Comparison of external joint torque and end-effector displacement under soft and stiff control.

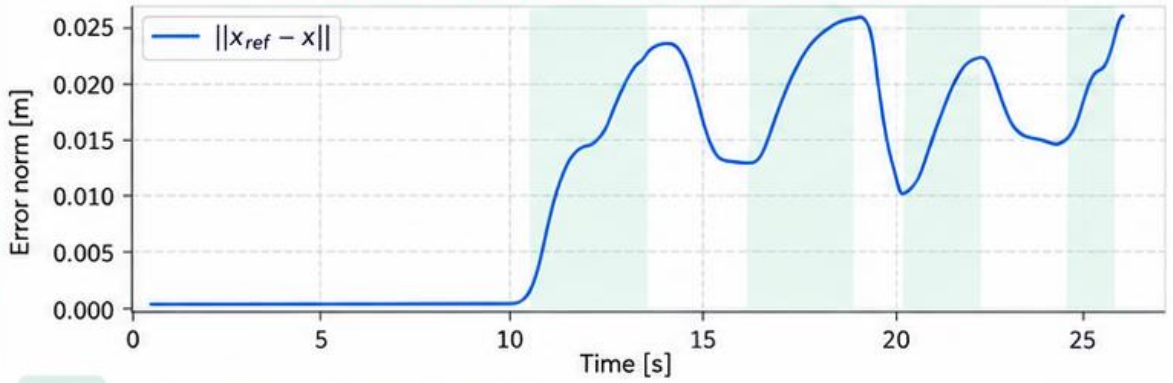
SOFT CONTROLLER

External Joint Torque Norm



Shaded areas: Human push intervals

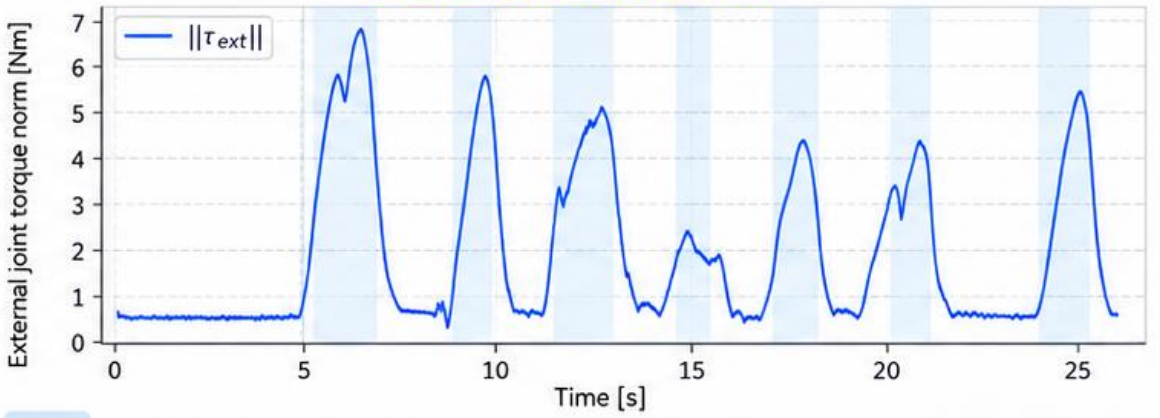
End-Effector Displacement from Reference



Shaded areas: Human push intervals

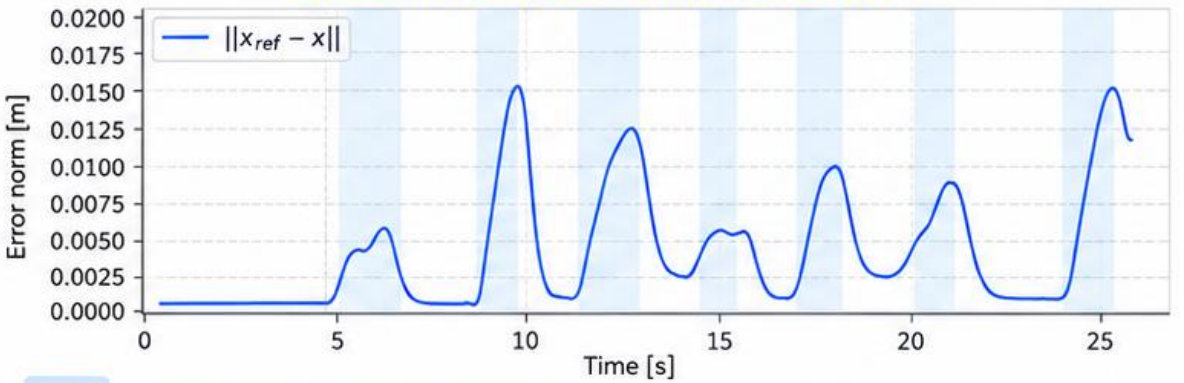
STIFF CONTROLLER

External Joint Torque Norm



Shaded areas: Human push intervals

End-Effector Displacement from Reference

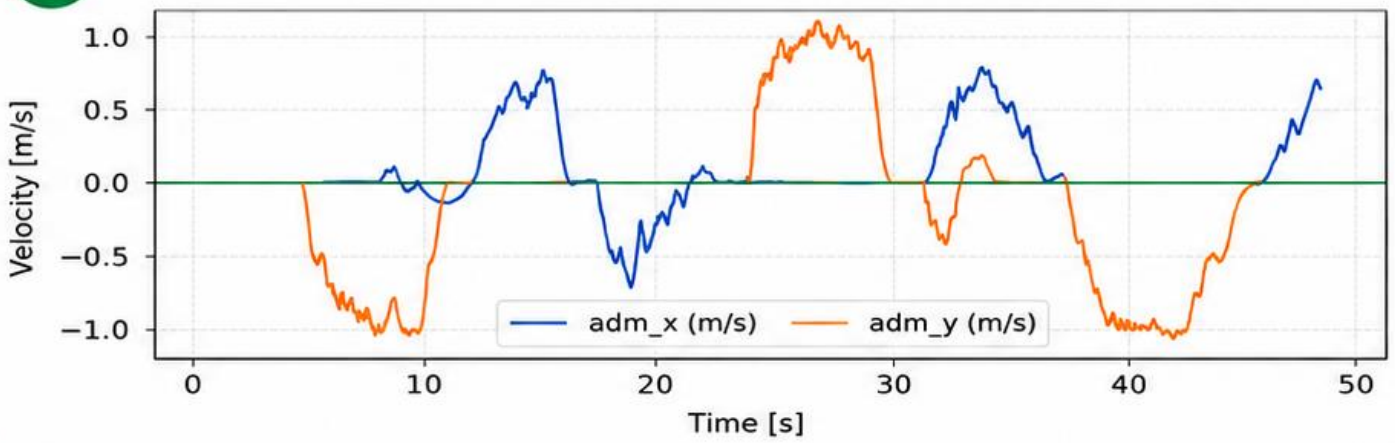


Shaded areas: Human push intervals

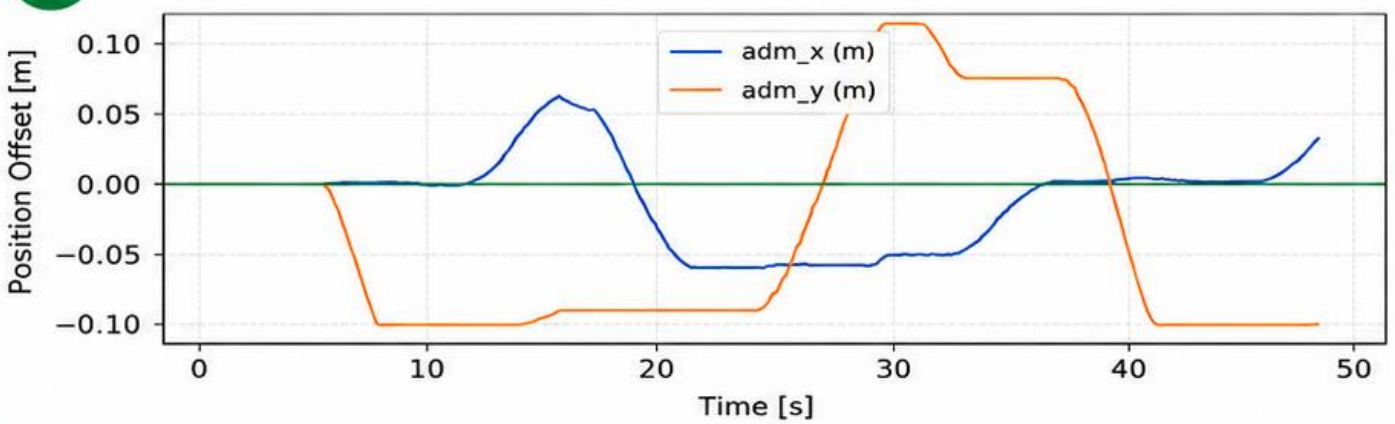
Hand-Guiding and Lead-Through

Objective: Demonstrate that a human can physically guide the robot using applied force.

Admittance-Generated Velocity



Admittance-Generated Position Offset

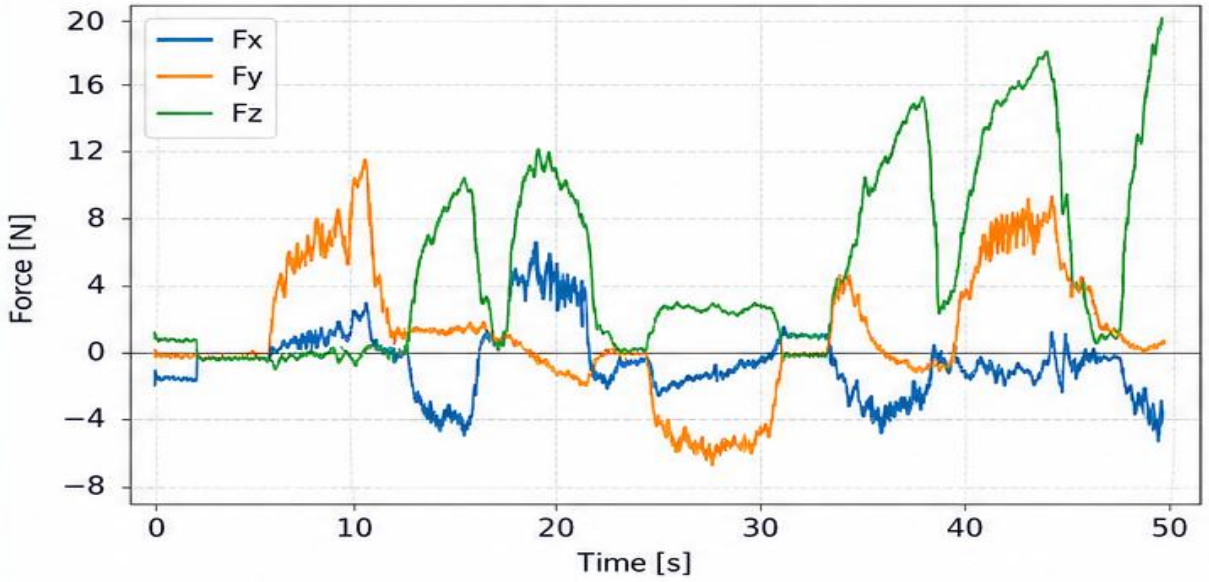




Hand-Guiding and Lead-Through

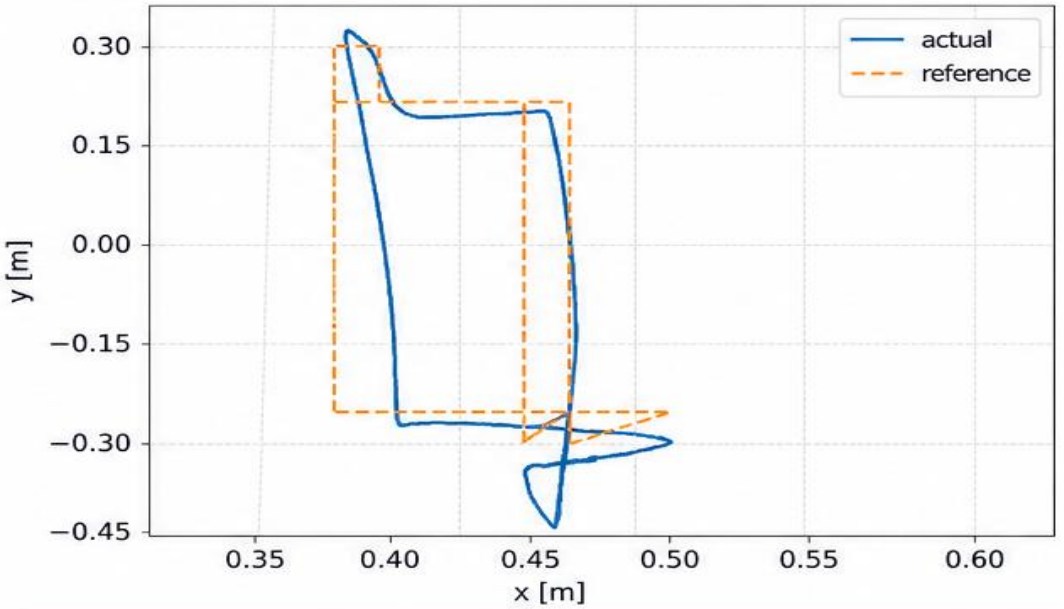
Objective: Demonstrate that a human can physically guide the robot using applied force.

Raw x/y/z External Force Components



- External forces measured at the end-effector while the human guides the robot.

Hand-Guided Path in x–y Plane

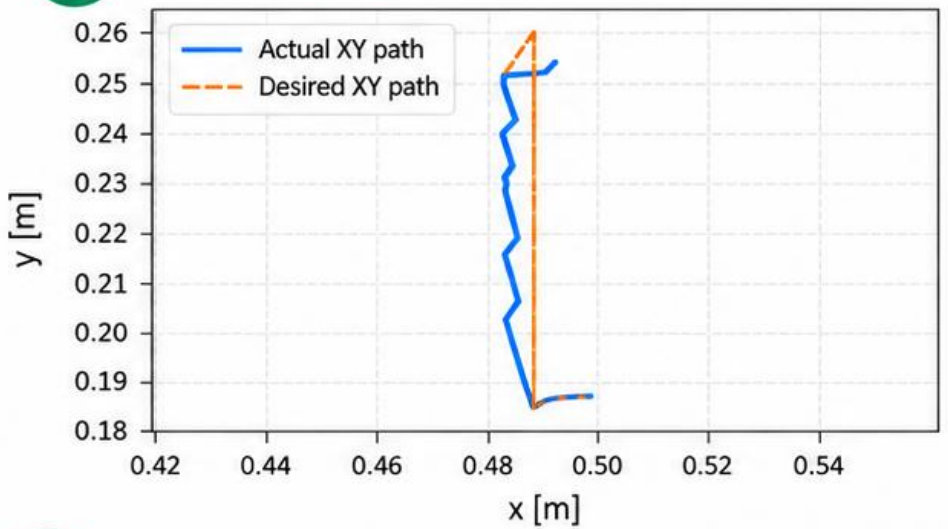


- The end-effector follows the desired path in the x–y plane, demonstrating lead-through capability.

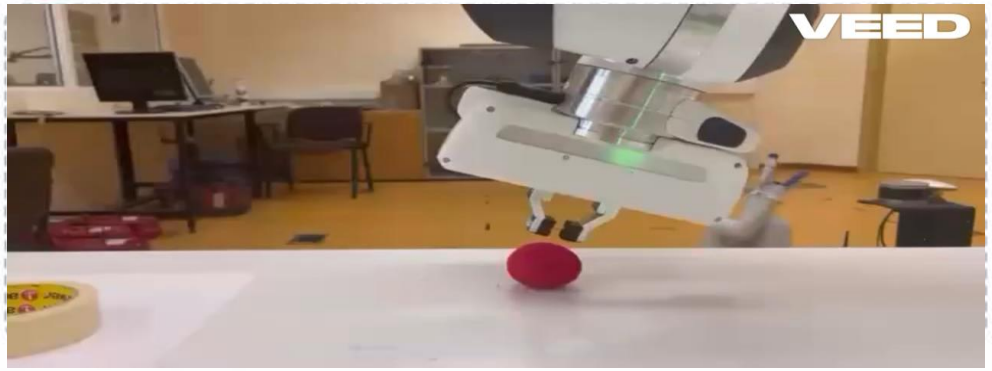
Hybrid Position/Force Interaction with a Soft Ball

Objective: Demonstrate the hybrid interaction capability through position regulation and force feedback using a soft ball as the contact object.

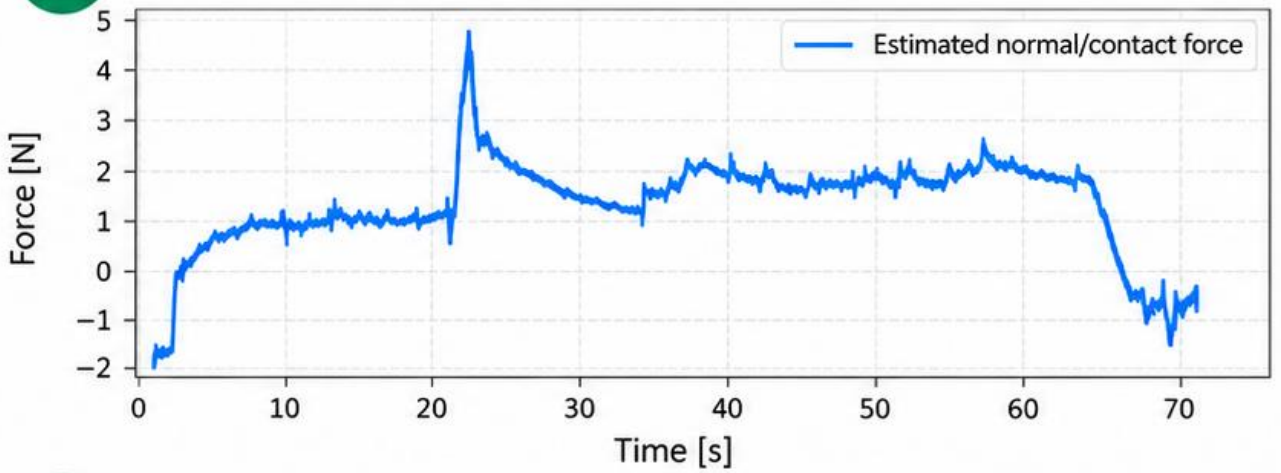
XY Dragging Path



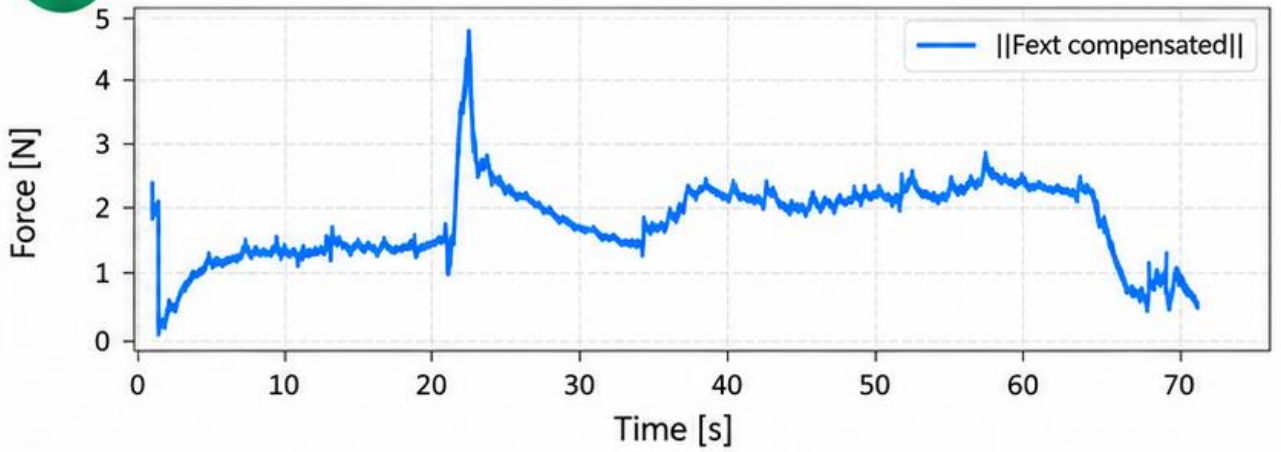
Video Demonstration



Estimated Z Contact Force

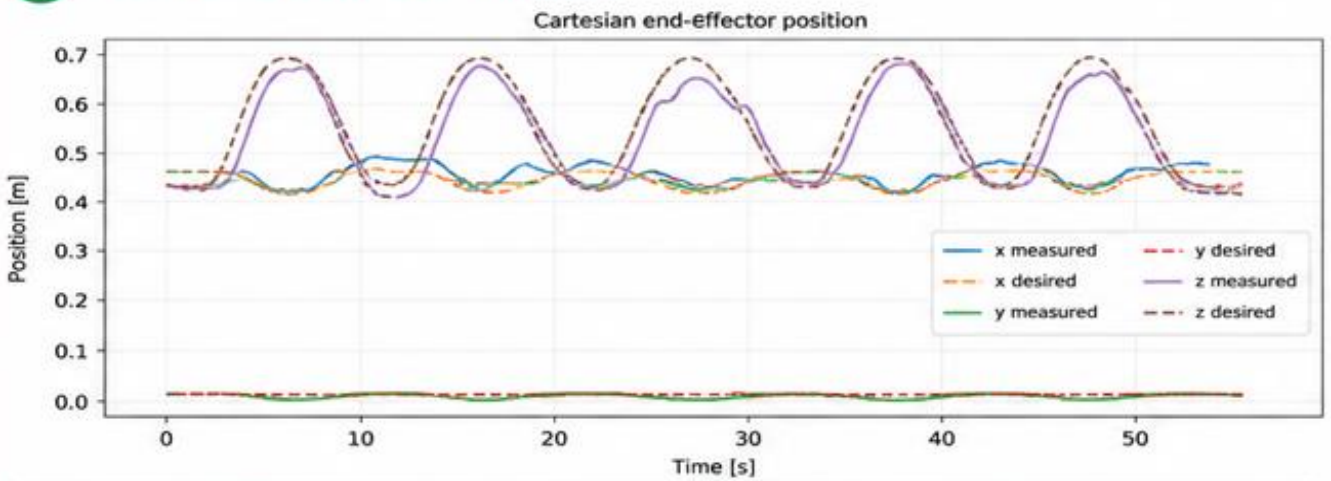


Bias-Compensated External Force Norm



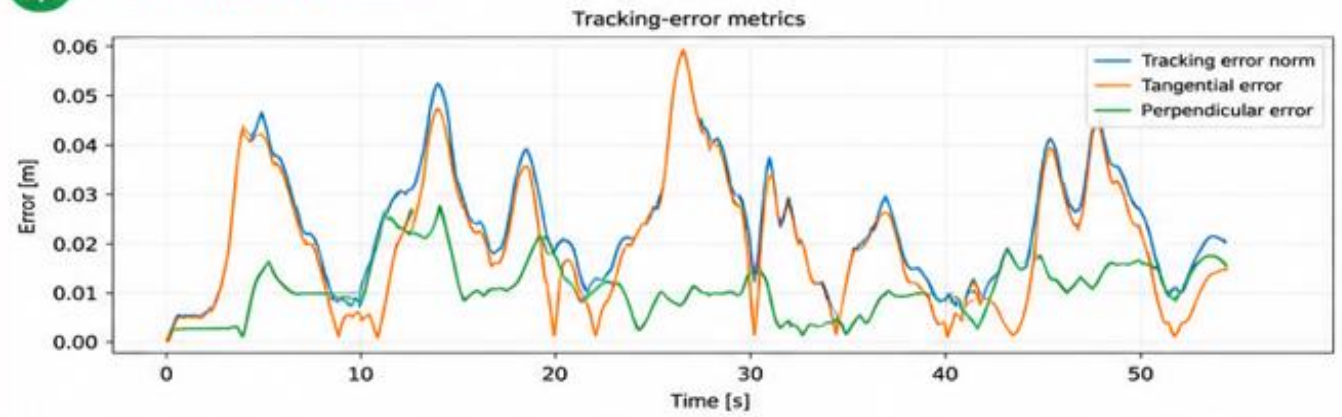
Resistance-Aware Assist-as-Needed Shoulder Rehabilitation

Cartesian End-Effector Position

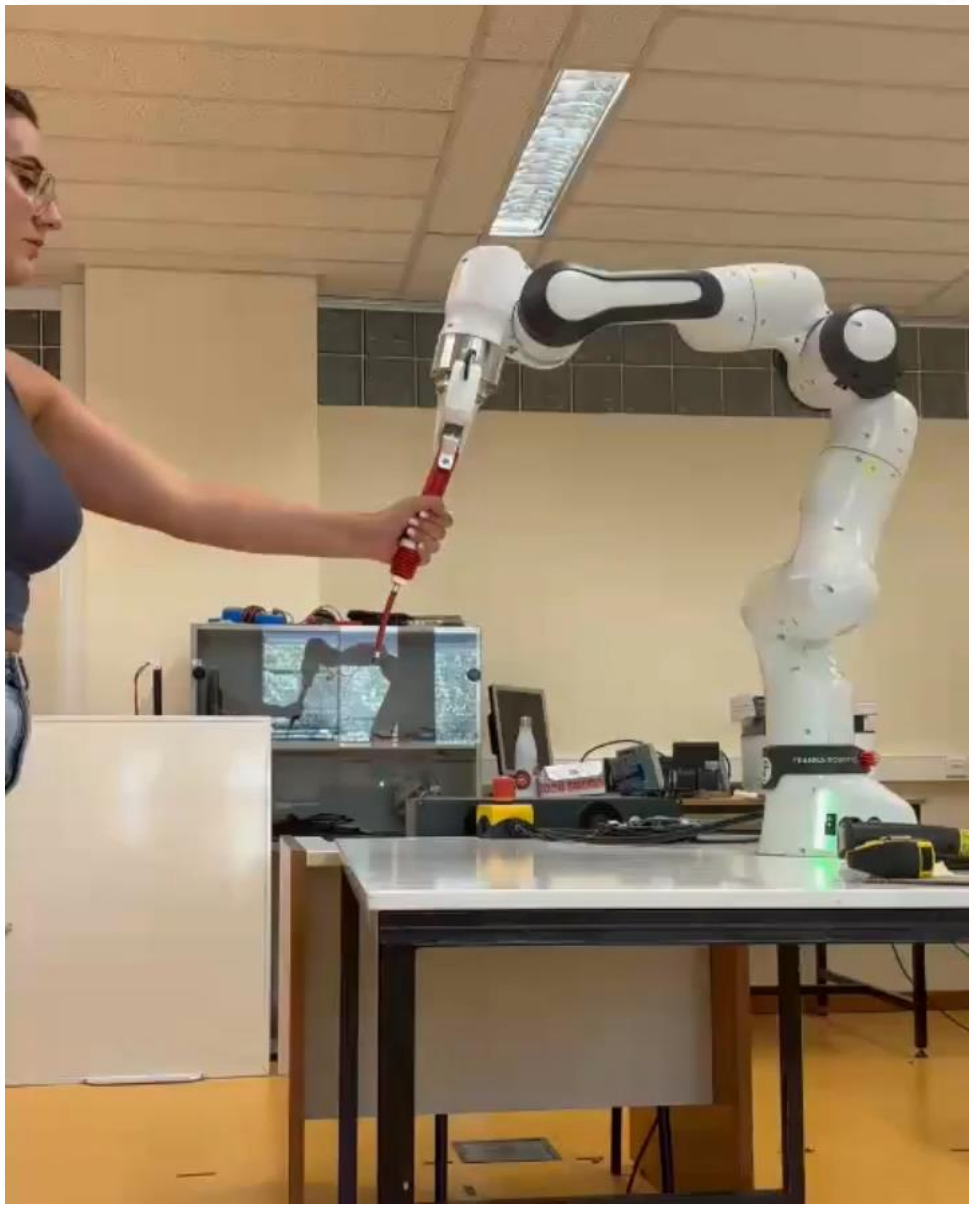


- The measured end-effector position closely follows the desired trajectory in all Cartesian axes.

Tracking-Error Metrics



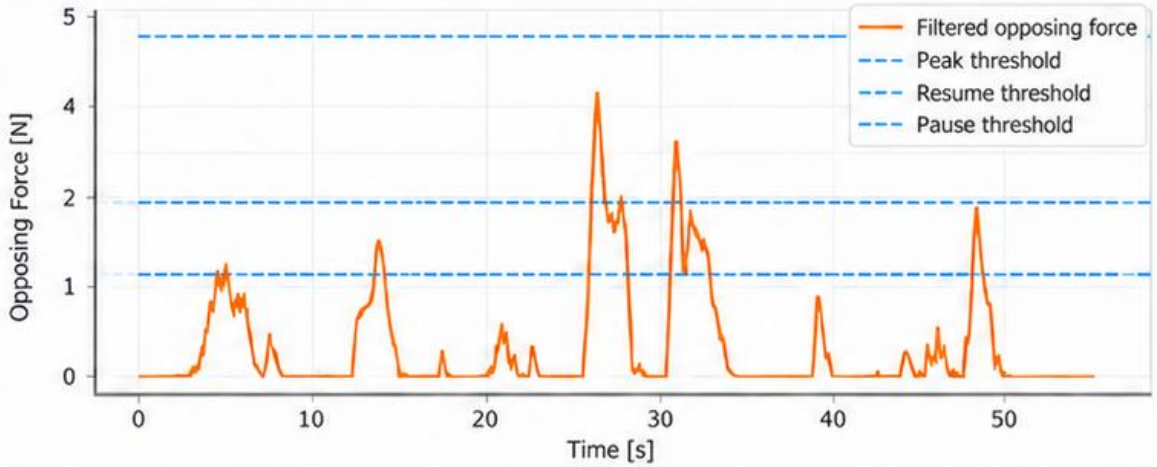
- Low tracking errors with small tangential and perpendicular components indicate accurate and stable guidance.



Resistance-Aware Assist-as-Needed Shoulder Rehabilitation

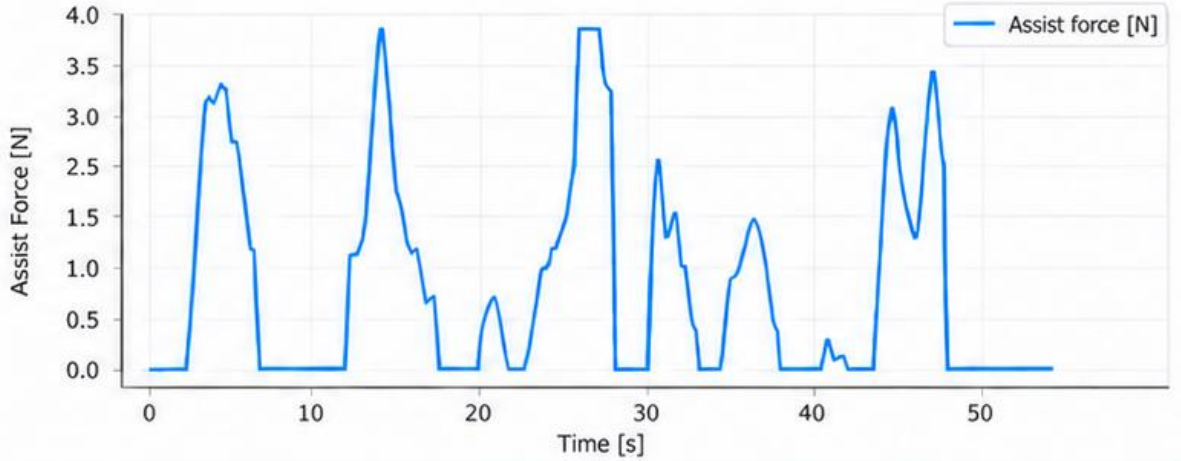
Objective: Adapt assistance based on the user's interaction resistance to provide safe, effective, and personalized shoulder rehabilitation.

Resistance Detection and Controller Thresholds



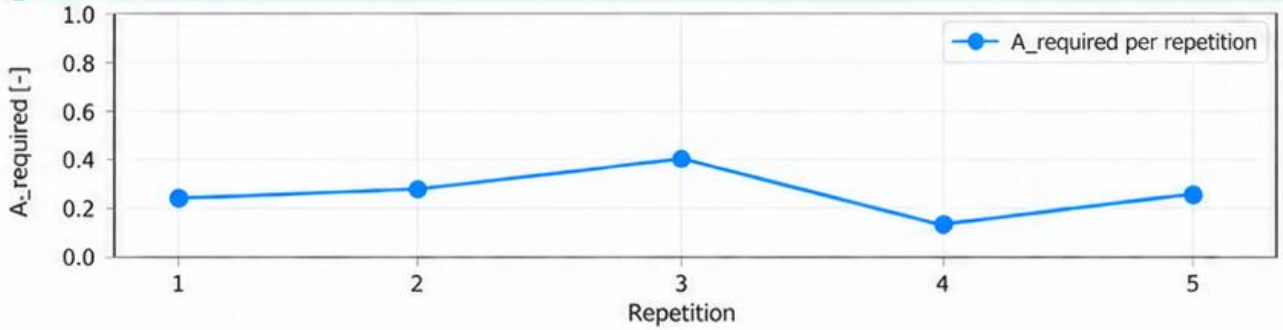
- Opposing force is monitored and compared against thresholds to determine when to increase, hold, or reduce assistance.

Assist as-needed Force



- Assistance is provided only when needed and adapts to the user's resistance. Cumulative normalized assistance reflects the total support delivered over time.

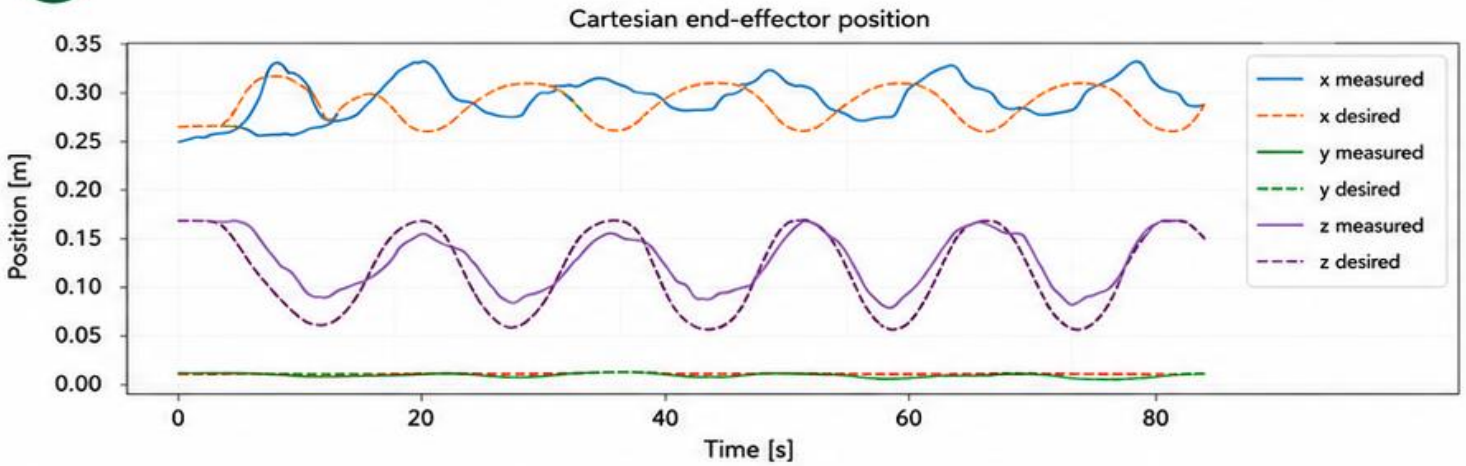
Required Assistance Across Repetitions



- Required assistance generally decreases across repetitions, indicating improved user performance and adaptation.
- The system provides just the right amount of help—no more, no less.

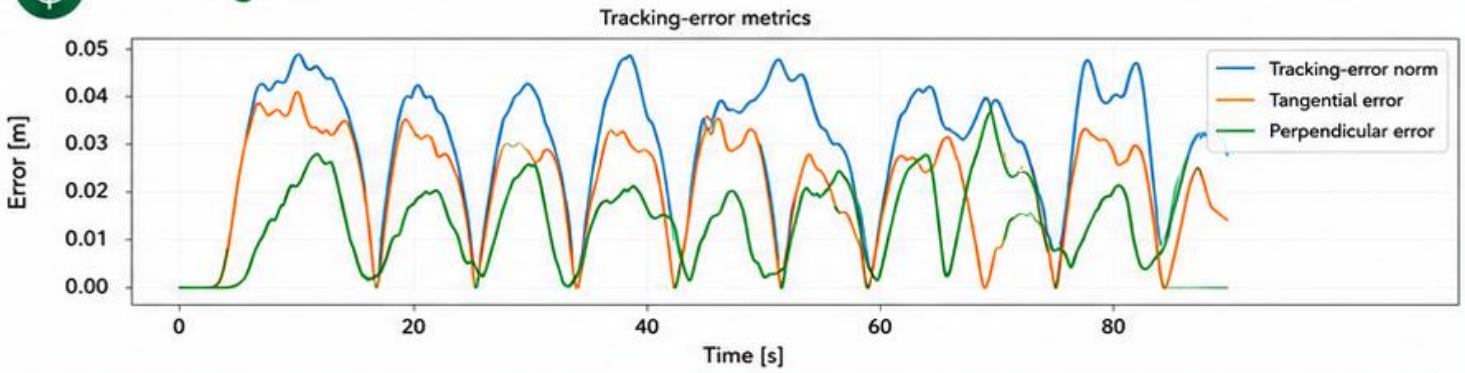
Resistance-Aware Assist-as-Needed Elbow Rehabilitation

Cartesian End-Effector Position

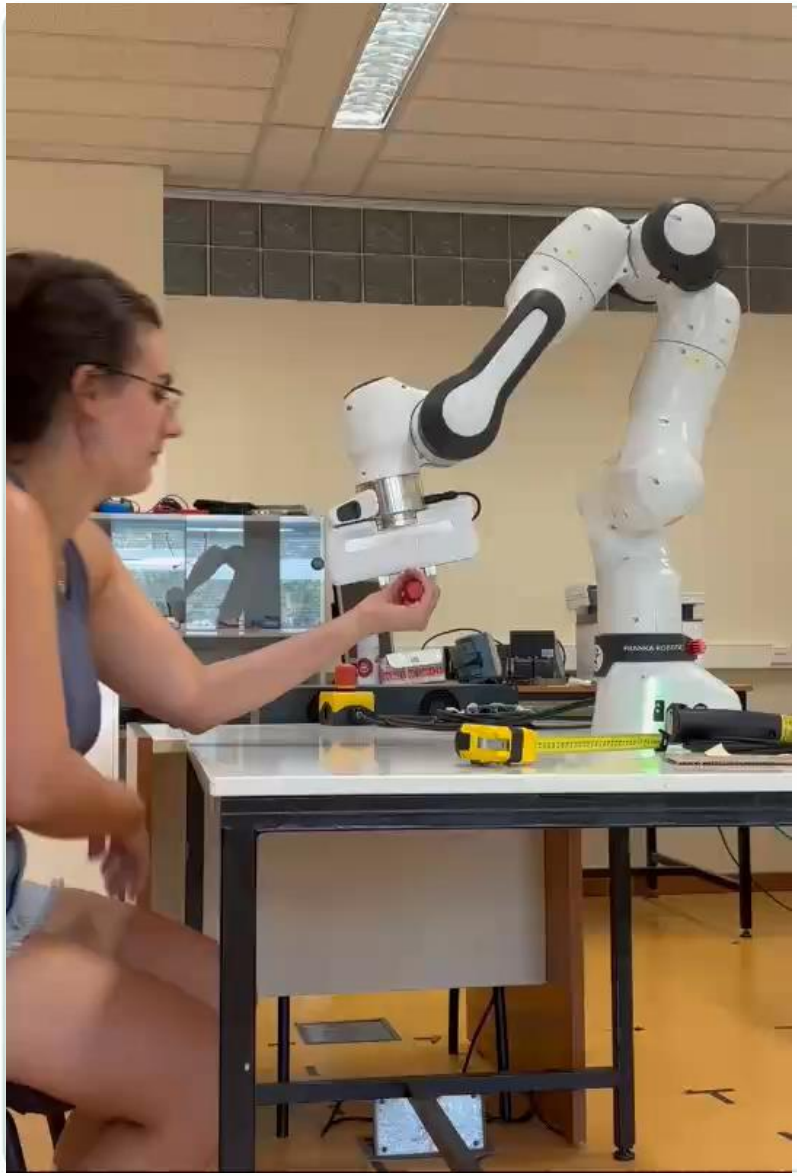


- The measured end-effector position closely follows the desired trajectory in all Cartesian axes.

Tracking-Error Metrics



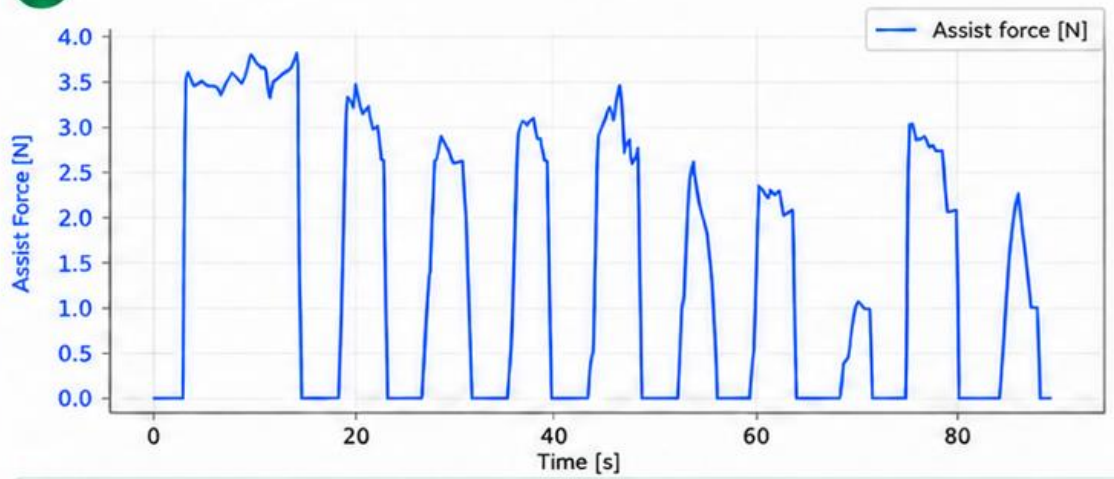
- Small tracking errors with low tangential and perpendicular components indicate accurate and stable control.



Resistance-Aware Assist-as-Needed Elbow Rehabilitation

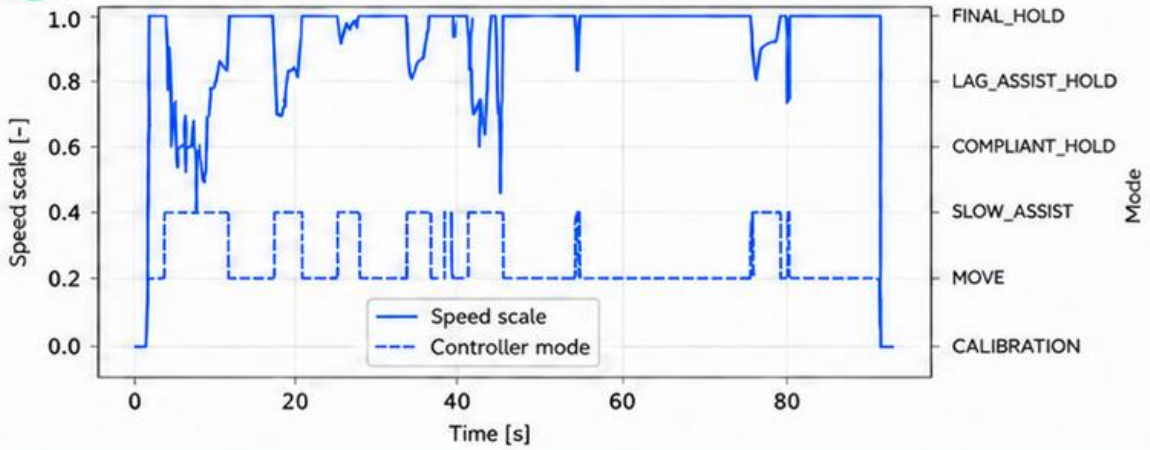
Objective: Provide adaptive assistance for elbow rehabilitation by adjusting support based on the user's resistance and promoting active participation.

Assist as needed Force



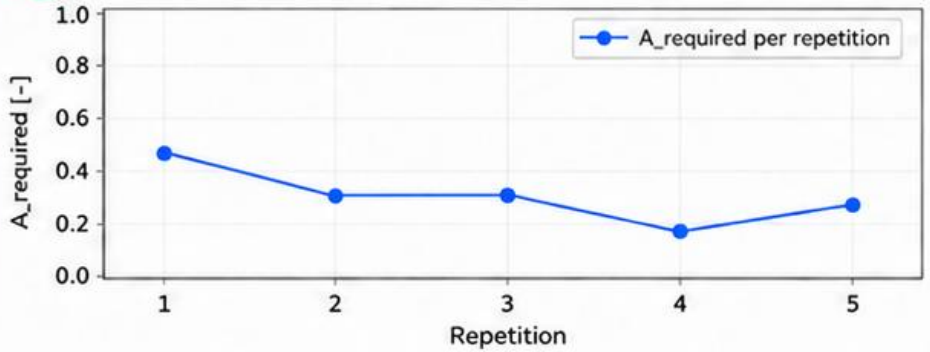
- Assistance is provided only when needed and decreases as the user becomes more capable. The cumulative normalized assistance reflects total support delivered over time.

Adaptive Progression and Controller Mode



- The controller adapts both speed and mode based on resistance: Calibration → Move → Slow Assist → Compliant Hold → Lag Assist Hold → Final Hold.

Required Assistance Across Repetitions



- Required assistance decreases across repetitions, indicating improved user performance and adaptation.
- The system provides just the right amount of support—no more, no less.

Conclusions

01 OK

FR3 platform

FR3 is a suitable force-aware collaborative platform due to torque sensing and FCI access.

02 F

Interaction controllers

Impedance, admittance, and hybrid force/position control provide complementary interaction behaviors.

03

Force tracking

Accurate force tracking requires bias compensation, conservative gains, damping, and systematic tuning.

04

Future work

Vision-based localization, adaptive impedance, automatic tuning, multi-contact recognition, and physiotherapy-oriented trials.



Thank you

Questions?

Force Control | Safety | Human Interaction

AI was used for the preparation of this presentation